

Competitive Programming

Section 36 Editorial: Fenwick Trees

Competitive Programming Course

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1 Editorial overview

Problem	Main idea
A. Point Add Range Sum	See full algorithm and code below.
B. Inversion Count	See full algorithm and code below.
C. Range Add Point Query	See full algorithm and code below.

2 A. Point Add Range Sum

2.1 Restatement

Maintain an array. Operation 1 $i\ x$ adds x to $a[i]$. Operation 2 $l\ r$ asks for the sum of $a[l..r]$.

2.2 Algorithm

Identify the operation that must be fast, store exactly the aggregate information needed for that operation, and update the affected logarithmic frontier. The lesson for this section explains the invariant in detail; the implementation below is a direct transcription of that invariant.

For this problem, every query is processed in the order given except offline queries explicitly marked as offline. The answer is emitted immediately when the operation asks for one.

2.3 Correctness proof

The maintained structure stores the exact summary promised by the lesson: prefix blocks for Fenwick trees, segment aggregates for segment trees, power-of-two jumps for static structures, or the active frontier for heap/Mo-style processing. Each update changes precisely the summaries whose represented set contains the changed element or interval. Each query decomposes the requested object into summaries that are disjoint, idempotently overlapping, or maintained as the current active window. The merge formula for those summaries is exactly the mathematical definition of the requested answer, so every printed value is correct. $\square$

2.4 Complexity

The optimized operations are logarithmic unless the lesson says the structure is static or offline. Memory is linear or $O(n \log n)$ for sparse-table/binary-lifting preprocessing.

2.5 Full C++ solution

```
#include <bits/stdc++.h>
using namespace std;

struct Fenwick {
    int n;
    vector<long long> bit;
```

```

Fenwick(int n) : n(n), bit(n + 1, 0) {}
void add(int i, long long delta) {
    for (; i <= n; i += i & -i) bit[i] += delta;
}
long long sum(int i) const {
    long long res = 0;
    for (; i > 0; i -= i & -i) res += bit[i];
    return res;
}
long long range_sum(int l, int r) const {
    return sum(r) - sum(l - 1);
}
};

int main() {
    ios::sync_with_stdio(false);
    cin.tie(nullptr);
    int n, q;
    if (!(cin >> n >> q)) return 0;
    Fenwick bit(n);
    for (int i = 1; i <= n; i++) {
        long long x;
        cin >> x;
        bit.add(i, x);
    }
    while (q--) {
        int typ;
        cin >> typ;
        if (typ == 1) {
            int i;
            long long x;
            cin >> i >> x;
            bit.add(i, x);
        } else {
            int l, r;
            cin >> l >> r;
            cout << bit.range_sum(l, r) << '\n';
        }
    }
    return 0;
}

```

2.6 Full Python solution

```

import sys

class Fenwick:
    def __init__(self, n):

```

```

        self.n = n
        self.bit = [0] * (n + 1)
    def add(self, i, delta):
        while i <= self.n:
            self.bit[i] += delta
            i += i & -i
    def sum(self, i):
        res = 0
        while i > 0:
            res += self.bit[i]
            i -= i & -i
        return res
    def range_sum(self, l, r):
        return self.sum(r) - self.sum(l - 1)

def main():
    data = list(map(int, sys.stdin.buffer.read().split()))
    if not data:
        return
    n, q = data[0], data[1]
    a = data[2:2 + n]
    bit = Fenwick(n)
    for i, x in enumerate(a, 1):
        bit.add(i, x)
    idx = 2 + n
    out = []
    for _ in range(q):
        typ = data[idx]
        idx += 1
        if typ == 1:
            i, x = data[idx], data[idx + 1]
            idx += 2
            bit.add(i, x)
        else:
            l, r = data[idx], data[idx + 1]
            idx += 2
            out.append(str(bit.range_sum(l, r)))
    print("\n".join(out))

if __name__ == "__main__":
    main()

```

3 B. Inversion Count

3.1 Restatement

Given an array, count pairs $i < j$ with $a[i] > a[j]$. Values may repeat and may be large.

3.2 Algorithm

Identify the operation that must be fast, store exactly the aggregate information needed for that operation, and update the affected logarithmic frontier. The lesson for this section explains the invariant in detail; the implementation below is a direct transcription of that invariant.

For this problem, every query is processed in the order given except offline queries explicitly marked as offline. The answer is emitted immediately when the operation asks for one.

3.3 Correctness proof

The maintained structure stores the exact summary promised by the lesson: prefix blocks for Fenwick trees, segment aggregates for segment trees, power-of-two jumps for static structures, or the active frontier for heap/Mo-style processing. Each update changes precisely the summaries whose represented set contains the changed element or interval. Each query decomposes the requested object into summaries that are disjoint, idempotently overlapping, or maintained as the current active window. The merge formula for those summaries is exactly the mathematical definition of the requested answer, so every printed value is correct. $\square$

3.4 Complexity

The optimized operations are logarithmic unless the lesson says the structure is static or offline. Memory is linear or $O(n \log n)$ for sparse-table/binary-lifting preprocessing.

3.5 Full C++ solution

```
#include <bits/stdc++.h>
using namespace std;

struct Fenwick {
    int n;
    vector<int> bit;
    Fenwick(int n) : n(n), bit(n + 1, 0) {}
};
```

```

void add(int i, int delta) {
    for (; i <= n; i += i & -i) bit[i] += delta;
}

int sum(int i) const {
    int res = 0;
    for (; i > 0; i -= i & -i) res += bit[i];
    return res;
}

};

int main() {
    ios::sync_with_stdio(false);
    cin.tie(nullptr);
    int n;
    if (!(cin >> n)) return 0;
    vector<long long> a(n), vals;
    for (long long &x : a) {
        cin >> x;
        vals.push_back(x);
    }
    sort(vals.begin(), vals.end());
    vals.erase(unique(vals.begin(), vals.end()), vals.end());
    Fenwick bit((int)vals.size());
    long long inv = 0;
    for (int i = 0; i < n; i++) {
        int rank = int(lower_bound(vals.begin(), vals.end(), a[i]) - vals.begin()) + 1;
        inv += i - bit.sum(rank);
        bit.add(rank, 1);
    }
    cout << inv << '\n';
    return 0;
}

```

3.6 Full Python solution

```

import sys

class Fenwick:
    def __init__(self, n):
        self.n = n
        self.bit = [0] * (n + 1)
    def add(self, i, delta):
        while i <= self.n:
            self.bit[i] += delta
            i += i & -i
    def sum(self, i):
        res = 0
        while i > 0:

```

```

        res += self.bit[i]
        i -= i & -i
    return res

def main():
    data = list(map(int, sys.stdin.buffer.read().split()))
    if not data:
        return
    n = data[0]
    a = data[1:1 + n]
    values = {x: i + 1 for i, x in enumerate(sorted(set(a)))}
    bit = Fenwick(len(values))
    inv = 0
    seen = 0
    for x in a:
        rank = values[x]
        inv += seen - bit.sum(rank)
        bit.add(rank, 1)
        seen += 1
    print(inv)

if __name__ == "__main__":
    main()

```

4 C. Range Add Point Query

4.1 Restatement

Maintain an array. Operation 1 $l\ r\ x$ adds x to every element in $[l, r]$. Operation 2 i asks for the current value of $a[i]$.

4.2 Algorithm

Identify the operation that must be fast, store exactly the aggregate information needed for that operation, and update the affected logarithmic frontier. The lesson for this section explains the invariant in detail; the implementation below is a direct transcription of that invariant.

For this problem, every query is processed in the order given except offline queries explicitly marked as offline. The answer is emitted immediately when the operation asks for one.

4.3 Correctness proof

The maintained structure stores the exact summary promised by the lesson: prefix blocks for Fenwick trees, segment aggregates for segment trees, power-of-two jumps for static structures, or the active frontier for heap/Mo-style processing. Each update changes precisely the summaries whose represented set contains the changed element or interval. Each query decomposes the requested object into summaries that are disjoint, idempotently overlapping, or maintained as the current active window. The merge formula for those summaries is exactly the mathematical definition of the requested answer, so every printed value is correct. $\square$

4.4 Complexity

The optimized operations are logarithmic unless the lesson says the structure is static or offline. Memory is linear or $O(n \log n)$ for sparse-table/binary-lifting preprocessing.

4.5 Full C++ solution

```
#include <bits/stdc++.h>
using namespace std;

struct Fenwick {
    int n;
    vector<long long> bit;
```

```

Fenwick(int n) : n(n), bit(n + 2, 0) {}
void add(int i, long long delta) {
    for (; i <= n; i += i & -i) bit[i] += delta;
}
long long sum(int i) const {
    long long res = 0;
    for (; i > 0; i -= i & -i) res += bit[i];
    return res;
}
};

int main() {
    ios::sync_with_stdio(false);
    cin.tie(nullptr);
    int n, q;
    if (!(cin >> n >> q)) return 0;
    Fenwick bit(n + 1);
    long long prev = 0;
    for (int i = 1; i <= n; i++) {
        long long x;
        cin >> x;
        bit.add(i, x - prev);
        prev = x;
    }
    while (q--) {
        int typ;
        cin >> typ;
        if (typ == 1) {
            int l, r;
            long long x;
            cin >> l >> r >> x;
            bit.add(l, x);
            bit.add(r + 1, -x);
        } else {
            int i;
            cin >> i;
            cout << bit.sum(i) << '\n';
        }
    }
    return 0;
}

```

4.6 Full Python solution

```

import sys

class Fenwick:
    def __init__(self, n):

```

```

        self.n = n
        self.bit = [0] * (n + 2)
    def add(self, i, delta):
        while i <= self.n:
            self.bit[i] += delta
            i += i & -i
    def sum(self, i):
        res = 0
        while i > 0:
            res += self.bit[i]
            i -= i & -i
        return res

def main():
    data = list(map(int, sys.stdin.buffer.read().split()))
    if not data:
        return
    n, q = data[0], data[1]
    a = data[2:2 + n]
    bit = Fenwick(n + 1)
    prev = 0
    for i, x in enumerate(a, 1):
        bit.add(i, x - prev)
        prev = x
    idx = 2 + n
    out = []
    for _ in range(q):
        typ = data[idx]
        idx += 1
        if typ == 1:
            l, r, x = data[idx], data[idx + 1], data[idx + 2]
            idx += 3
            bit.add(l, x)
            bit.add(r + 1, -x)
        else:
            i = data[idx]
            idx += 1
            out.append(str(bit.sum(i)))
    print("\n".join(out))

if __name__ == "__main__":
    main()

```

5 Testing discussion

The fixed samples cover visible behavior. The random tests generate many small operation sequences and compute expected outputs from the reference Python implementation, then the same expected files are used against both the C++ and Python submissions.

6 Problem-specific upsolve notes

6.1 A. Point Add Range Sum

Constraint implication and trap. With a long sequence of updates and queries, scanning $[1, r]$ per query is quadratic in the worst case. The update is an addition, not an assignment: call `add(i, x)`, never overwrite the old Fenwick contribution.

Walk-through. From $[1, 2, 3, 4]$, query $[2, 4]$ is `prefix(4)-prefix(1)=9`. After `add(3, 10)`, the same query is 19. The tree's prefix invariant proves the subtraction removes exactly positions before 1.

```
build: for i=1..n: add(i, a[i])
type 1 i x: add(i, x)
type 2 l r: print prefix(r)-prefix(l-1)
```

Every `bit[j]` stores its lowbit block sum. Prefix blocks partition $[1, r]$ and $[1, l-1]$, so their difference is exactly $[l, r]$. Build is $O(n \log n)$ with these calls and each operation is $O(\log n)$; memory is $O(n)$. In both C++ and Python, use 64-bit-capable sums. Test $l=1$, $l=r$, negative delta, and a query after several additions to one position.

6.2 B. Inversion Count

Observation. Values can be large or negative, so compress them to ordered ranks. At index i , $i - \text{prefix}(\text{rank})$ is the number of earlier values that are strictly greater; using `prefix(rank-1)` would incorrectly count equals.

For $[3, 1, 2, 5, 4]$, the contributions are $0, 1, 1, 0, 1$, total 3. The Fenwick invariant counts exactly the processed values of each rank, and compression preserves $<$, proving the accumulated contributions count every inversion once when its right endpoint arrives. Complexity is $O(n \log n)$ time and $O(n)$ memory. C++ needs `long long inv`; Python needs no special numeric type. Test sorted, reverse-sorted, all-equal, and values outside 32-bit range.

```
compress values to ranks
for i from 0 to n-1:
    answer += i - prefix(rank[a[i]])
    add(rank[a[i]], 1)
```

6.3 C. Range Add Point Query

Observation. Store `diff`, not the logical values. A range add changes only `diff[l]` and `diff[r+1]`; a point value is the prefix sum of differences. For `[10,20,30,40]`, add 5 on `[2,3]`: prefix at 3 changes from 30 to 35, but prefix at 4 returns to 40 because of `diff[4]=-5`.

The difference identity proves that every position in `[1,r]` sees `+x` once and every other position sees net zero. Each update and point query is $O(\log n)$, with $O(n)$ memory. Allocate an extra usable index for `r+1`. In C++/Python, initialize by inserting `a[i]-a[i-1]`, not `a[i]`. Test a full-range add, `r=n`, and a negative update.

7 Stress-testing workflow

For A and C, keep a slow list and mutate it literally; for B, use a double loop on arrays of length at most eight. Compare after every random operation, then run `CP_TARGET=solution python3 sections/36_fenwick_trees/check.py`. Carry forward: Fenwick trees are prefix structures, and coordinate compression or a difference transform changes what that prefix means.